Supplementary Material for Decoys That Do Work: Evaluating Executed Cover for LLM Agent Communication Privacy

This supplement gives the separate state-control cohort (S1), the live TLS-record study (S2), record-only analysis (S3), and reproduction scope (S4). The accompanying evidence archive preserves the source-inspection material and historical pilots with their original results. Reference numbers are shared with the main paper.

S1. LLM-planned executed cover

The experiment measures traffic recognition, requested-task completion and full business-state integrity on the same executions. It uses six synthetic ticket actions and six policies: no cover, proxy discard, read cover, hidden isolated execution, live writes and visible isolated execution. All policies share fixtures and frozen random cover draws. The earlier reference-program replay remains a separate artifact; its fixed-position and one-to-four-decoy sweeps are not LLM results.

S1.1. Population, planner and cover semantics

Each of two fixed-weight models, Qwen3:8B and Llama3.1:8B, runs 300 fixtures per condition: 120 training, 60 validation and 120 test, balanced across six requested actions. This gives 3,600 episodes, including 1,440 test executions of 120 unique paired test fixtures. Request phrasings recur across splits; this is fixture holdout, not unseen-template transfer. The seed pairs fixtures and decoy schedules, and condition order rotates. Two workers collect each model cohort on one host.

LangGraph [56] alternates model decisions with actual tool calls. The controller fetches the ticket; required checks constrain action admissibility without naming the correct action or object. The model chooses check order, requested business operation, order identifier, recovery and finish, up to ten decision steps. Temperature is zero, context 4,096 and maximum generated decision length 96 tokens; the fixture supplies the model seed. Fixed manifests and weight blobs match before and after collection.

The fixture contains three orders, customer address, a compensation quota, escalation status and a ledger. Its intended reference is the initial state after exactly the requested legitimate action. Decoys deliberately include operations capable of collateral changes. A frozen schedule places one or two cover calls after the ticket or early honest calls; extra later planner calls do not add cover. Each cover cell triggers all 182 planned test decoys.

Proxy discard computes the response on a disposable copy, retaining no decoy state. Read cover queries state without changing it. Hidden isolated execution applies the decoy on a persistent clone; live write applies it to the primary fixture. Visible isolated execution uses the same clone and schedule, forwarding clearly marked gateway-cover requests and responses to the planner. These objects are not fabricated planner tool calls. In the other conditions, cover responses are hidden from the planner.

S1.2. Transport, observer and endpoints

Seven local MCP Streamable HTTP destinations use actual TLS 1.3/HTTP/2 sockets and a common `/mcp` path. Request bodies use a 4,096-byte padding quantum and responses 16,384 bytes. All seven sessions initialize before each episode and are reused within it. Total TLS cost includes those initializations; call-byte cost and attack features exclude them. The observer receives destination and bidirectional ciphertext byte counts in supplied episode and logical-call windows. It receives no plaintext, model traffic, physical timing, arm label, fixture identifier or private stream assignment. The original replay's variable HTTP-path artifact therefore does not carry over.

Recognition targets the requested action, retaining wrong or failed executions. Fixed features encode the first sixteen call destinations and ciphertext byte counts, per-destination totals and total calls/bytes. LR uses training-only standardization, $C = 1$ and a 3,000-iteration cap. RF uses 300 trees with fixed seed. Validation balanced accuracy selects LR or RF within each model/condition, breaking ties toward LR. Both candidates, the no-cover-selected frozen attack and last/penultimate destination rules remain reported (Table S1). The schedule was frozen before formal collection. The analysis freeze at 17:05 UTC on 10 November 2026 included both position rules with zero terminal test episodes. A logged pre-test amendment at 17:11 handled missing penultimate calls using the training-majority class; it did not change the primary attacker.

Goal completion requires the requested action/object, satisfied prerequisites and the target postcondition. State integrity compares the full business state, including non-target orders, shared fields and the ledger, against the single

intended-action reference. The predicates differ: a reference-matching state can still fail the stricter action/object-history goal predicate. Conversely, a completed goal can leave collateral changes. Table S2 reports both and their completed-but-incorrect-state intersection.

S1.3. Paired inference and uncertainty

The analysis fixes five cover-minus-none contrasts plus visible-minus-hidden for each model. Each endpoint—recognition, completion and integrity—has its own twelve-comparison Holm family. Two-sided exact discordant-pair tests use paired fixture outcomes. Intervals use 10,000 paired fixture bootstrap draws stratified by requested action and condition on the fitted attackers and designed workload. They do not represent template, fit-selection or deployment uncertainty. Table S3 retains every prespecified comparison, including null contrasts.

All ten primary recognition contrasts versus no cover survive correction; the largest adjusted p is 3.08×10^{-8}. Live-write integrity reductions survive for both models (3.70×10^{-32} and 2.98×10^{-19}), while completion survives only for Qwen (.02197, versus 1 for Llama). Every primary visible-minus-hidden contrast has adjusted $p = 1$. These tests do not establish equivalence among cover policies or a privacy upper bound. In particular, Qwen's last-destination rule exceeds the validation-selected learner under cover.

S1.4. Planner paths, costs and limitations

Hidden and visible isolated runs have identical cover responses in all 120 paired fixtures per model. Their honest action sequences differ in 16 Qwen and 65 Llama pairs. Hidden-arm comparisons also vary: proxy versus hidden-isolated differs in one Qwen and fifteen Llama pairs; read versus hidden-isolated differs in zero and fifteen. Pairing fixes the input fixture and schedule, not the realized planner trajectory. These path differences limit mechanism attribution; no repeated-condition cohort estimates the planner noise floor. Mean prompt tokens rise from 1,668.34 to 1,969.49 for Qwen and from 2,251.48 to 2,675.95 for Llama when responses are visible, increases of 18.1% and 18.9%.

Cover call-byte ratios are 1.280–1.346; total TLS ratios are 1.131–1.149 because session initialization is charged to both conditions. Mean local wall times are 8.14–9.94 seconds per test episode, including setup, journaling and two-worker contention. These are local resource measurements. Test outputs contain no invalid decision JSON. Llama reaches the step limit once in no-cover, proxy, read and hidden-isolated conditions, twice in live-write and zero times in visible-isolated; Qwen never does. Every terminal case remains included.

All 3,600 formal episodes, 16,601 model request/response chains, 21,163 tool calls and 4,550 cover calls pass separate state, request and transport checks. Formal unresolved requests are zero. One earlier development request remains unknown, was not retried and is excluded from the formal cohort. The release supplies synthetic planner records, fixture states, lengths, predictions and replay scripts; it excludes credentials, model weights, private TLS keys and raw packet captures.

The adverse writes make the consequence measurable rather than estimating its production prevalence. The experiment is neither a reproduction of WRA's research decoys nor a measurement of physical-timing, model-link or WAN privacy. Llama's unprotected completion and state baselines are already imperfect. The case applies established state-aware evaluation to an adverse cover control, with traffic and cost recorded on those same executions. The direction of collateral harm is strongly predisposed by the chosen writes; the measured endpoint disagreement shows which failures completion omits. Visible-response averages remain inconclusive.

S1.5. Worked application of the evaluation procedure

This application fills the main paper's four-step procedure using saved observations. It adds no data, attacker selection or acceptance threshold.

1. Target and observations. The private target is the requested action among six balanced classes. The observer receives tool destinations and encrypted call-byte counts, with episode and call windows supplied. The model service and physical timestamps are outside the tested view. This original view has no model-link evaluation. The same executions receive a separate client-receipt-timing analysis in the preserved historical source archive; it retains supplied call windows.

2. Allowed execution effects. The synthetic application allows exactly the requested legitimate business action. The reference records its effect on the target and other orders, address, quota, escalation state and ledger. Goal completion additionally requires the correct action/object history and prerequisites. Exact state equality is appropriate to this isolated fixture with no legitimate concurrent updates. A contract can be derived by listing the requested operation's permitted changes, adding frame conditions for all other objects, and specifying quota and ledger updates allowed by that operation. Here the reference executor implements that rule once and the checker compares every field. In production, the contract must also permit legitimate concurrent updates and specify resource budgets; exact fixture equality alone would be inappropriate.

3. Paired measurements. The six conditions share fixtures and cover draws. Every test execution contributes to recognition, completion and state outcomes. LR/RF train on each condition; frozen and destination-position rules are also reported. The joint count of completion with an incorrect state is measured from the two predicates, rather than generally calculated by subtraction. Qwen's live-write arm has 102 completions, of which 95 fail the state contract; Llama has 76, of which 73 fail it. All 36 prespecified paired endpoint comparisons remain in Table S3 and its continuations. Tables S1–S2 give the condition-level outcomes.

TABLE S1. LLM-PLANNED EXECUTED COVER ON 120 PAIRED HELD-OUT FIXTURES PER CELL. THE PRIMARY LR/RF ATTACKER IS SELECTED ON VALIDATION DATA. ALL ATTACK COLUMNS COUNT CORRECT PREDICTIONS OUT OF 120; CHANCE IS 20/120. THE INTERVAL CONDITIONS ON THE FITTED ATTACKER AND SYNTHETIC FIXTURES.

Model	Cover	Frozen	LR	RF	Last	Penultimate	Selected BA [95% CI]
Qwen3:8B	No cover	118	118	117	116	14	0.983 [0.958, 1.000] (LR)
Qwen3:8B	Proxy discard	44	63	62	77	43	0.517 [0.425, 0.608] (RF)
Qwen3:8B	Read cover	42	59	62	77	42	0.517 [0.425, 0.608] (RF)
Qwen3:8B	Isolated, hidden	43	62	61	77	42	0.508 [0.417, 0.600] (RF)
Qwen3:8B	Live write	42	60	62	75	44	0.517 [0.425, 0.608] (RF)
Qwen3:8B	Isolated, visible	40	63	63	77	42	0.525 [0.433, 0.617] (RF)
Llama3.1:8B	No cover	113	113	110	79	64	0.942 [0.900, 0.975] (LR)
Llama3.1:8B	Proxy discard	54	81	79	67	61	0.658 [0.575, 0.733] (RF)
Llama3.1:8B	Read cover	57	78	72	67	62	0.600 [0.517, 0.683] (RF)
Llama3.1:8B	Isolated, hidden	56	81	82	67	60	0.683 [0.600, 0.758] (RF)
Llama3.1:8B	Live write	58	80	75	65	59	0.667 [0.583, 0.750] (LR)
Llama3.1:8B	Isolated, visible	58	79	81	68	64	0.675 [0.592, 0.750] (RF)

TABLE S2. TASK OUTCOMES AND MEASURED COSTS ON THE SAME 120 HELD-OUT FIXTURES. STATE INTEGRITY COMPARES THE ENTIRE BUSINESS STATE WITH THE SINGLE INTENDED-ACTION REFERENCE. BYTE RATIOS ARE MEANS OF PAIRED RATIOS; TOTAL TLS BYTES INCLUDE ALL SEVEN MCP INITIALIZATIONS, WHILE CALL BYTES EXCLUDE INITIALIZATION.

Model	Cover	Goal	State	Goal + state error	Call bytes ×	Total bytes ×	Model calls
Qwen3:8B	No cover	114/120	116/120	0/120	1.00	1.00	4.06
Qwen3:8B	Proxy discard	114/120	116/120	0/120	1.34	1.15	4.06
Qwen3:8B	Read cover	114/120	115/120	0/120	1.34	1.15	4.07
Qwen3:8B	Isolated, hidden	114/120	115/120	0/120	1.34	1.15	4.07
Qwen3:8B	Live write	102/120	7/120	95/120	1.34	1.15	4.08
Qwen3:8B	Isolated, visible	113/120	114/120	1/120	1.35	1.15	4.08
Llama3.1:8B	No cover	82/120	69/120	13/120	1.00	1.00	5.23
Llama3.1:8B	Proxy discard	80/120	68/120	13/120	1.29	1.14	5.31
Llama3.1:8B	Read cover	84/120	71/120	14/120	1.28	1.13	5.25
Llama3.1:8B	Isolated, hidden	83/120	70/120	13/120	1.28	1.13	5.25
Llama3.1:8B	Live write	76/120	3/120	73/120	1.29	1.14	5.35
Llama3.1:8B	Isolated, visible	83/120	73/120	11/120	1.29	1.14	5.33

4. Interpretation against the contract. Live writes fail the declared state requirement even when the target operation completes. Qwen's hidden isolated arm gives a descriptive inference–cost point: the original timing-free attack maximum is 77/120 versus 118/120 without cover, completion is 114/120 in both, integrity is 115/120 versus 116/120, and call bytes are 1.34 times baseline. This maximum summarizes the reported attacks; it does not replace the validation-selected primary attacker or its corrected tests. The added endpoint-timing removal attack instead reaches 112/120 for this isolated condition. No application acceptance tolerances or privacy upper bound were established; the original point cannot be used as its protection level against the expanded observer. The same record exposes what a deployment assessment still needs: broader observers, workload transfer, state tolerances and stronger schedule-aware attacks.

TABLE S3. GOAL COMPLETION: EVERY PRESPECIFIED PAIRED CONTRAST. DIFFERENCES AND CONDITIONAL INTERVALS ARE PERCENTAGE POINTS. THIS ENDPOINT HAS ONE TWELVE-COMPARISON HOLM FAMILY.

Model	Condition minus reference	Difference	Conditional 95% CI	Raw p	Holm p
Qwen3:8B	Proxy – No cover	0.00	[0.00, 0.00]	1.000	1.000
Qwen3:8B	Read – No cover	0.00	[0.00, 0.00]	1.000	1.000
Qwen3:8B	Hidden clone – No cover	0.00	[0.00, 0.00]	1.000	1.000
Qwen3:8B	Live write – No cover	-10.00	[-15.00, -5.00]	0.002	0.022
Qwen3:8B	Visible clone – No cover	-0.83	[-4.17, 2.50]	1.000	1.000
Qwen3:8B	Visible clone – Hidden clone	-0.83	[-4.17, 2.50]	1.000	1.000
Llama3.1:8B	Proxy – No cover	-1.67	[-5.00, 1.67]	0.625	1.000
Llama3.1:8B	Read – No cover	1.67	[-2.50, 5.83]	0.688	1.000
Llama3.1:8B	Hidden clone – No cover	0.83	[0.00, 2.50]	1.000	1.000
Llama3.1:8B	Live write – No cover	-5.00	[-10.83, 0.83]	0.210	1.000
Llama3.1:8B	Visible clone – No cover	0.83	[-6.67, 8.33]	1.000	1.000
Llama3.1:8B	Visible clone – Hidden clone	0.00	[-7.50, 7.50]	1.000	1.000

TABLE S4. REQUESTED-ACTION RECOGNITION: EVERY PRESPECIFIED PAIRED CONTRAST. DIFFERENCES AND CONDITIONAL INTERVALS ARE PERCENTAGE POINTS. THIS ENDPOINT HAS ONE TWELVE-COMPARISON HOLM FAMILY.

Model	Condition minus reference	Difference	Conditional 95% CI	Raw p	Holm p
Qwen3:8B	Proxy – No cover	-46.67	[-55.83, -37.50]	2.78×10^{-17}	3.05×10^{-16}
Qwen3:8B	Read – No cover	-46.67	[-55.00, -37.50]	2.78×10^{-17}	3.05×10^{-16}
Qwen3:8B	Hidden clone – No cover	-47.50	[-56.67, -38.33]	1.39×10^{-17}	1.67×10^{-16}
Qwen3:8B	Live write – No cover	-46.67	[-55.83, -37.50]	2.78×10^{-17}	3.05×10^{-16}
Qwen3:8B	Visible clone – No cover	-45.83	[-55.00, -36.67]	8.05×10^{-16}	6.44×10^{-15}
Qwen3:8B	Visible clone – Hidden clone	1.67	[-5.00, 8.33]	0.815	1.000
Llama3.1:8B	Proxy – No cover	-28.33	[-35.83, -20.83]	1.08×10^{-9}	5.59×10^{-9}
Llama3.1:8B	Read – No cover	-34.17	[-42.50, -25.83]	9.09×10^{-13}	6.37×10^{-12}
Llama3.1:8B	Hidden clone – No cover	-25.83	[-34.17, -18.33]	9.31×10^{-10}	5.59×10^{-9}
Llama3.1:8B	Live write – No cover	-27.50	[-35.83, -19.17]	1.02×10^{-8}	3.07×10^{-8}
Llama3.1:8B	Visible clone – No cover	-26.67	[-35.00, -19.17]	4.07×10^{-9}	1.63×10^{-8}
Llama3.1:8B	Visible clone – Hidden clone	-0.83	[-8.33, 6.67]	1.000	1.000

TABLE S5. BUSINESS-STATE INTEGRITY: EVERY PRESPECIFIED PAIRED CONTRAST. DIFFERENCES AND CONDITIONAL INTERVALS ARE PERCENTAGE POINTS. THIS ENDPOINT HAS ONE TWELVE-COMPARISON HOLM FAMILY.

Model	Condition minus reference	Difference	Conditional 95% CI	Raw p	Holm p
Qwen3:8B	Proxy – No cover	0.00	[0.00, 0.00]	1.000	1.000
Qwen3:8B	Read – No cover	-0.83	[-2.50, 0.00]	1.000	1.000
Qwen3:8B	Hidden clone – No cover	-0.83	[-2.50, 0.00]	1.000	1.000
Qwen3:8B	Live write – No cover	-90.83	[-95.00, -85.83]	3.08×10^{-33}	3.70×10^{-32}
Qwen3:8B	Visible clone – No cover	-1.67	[-5.83, 1.67]	0.688	1.000
Qwen3:8B	Visible clone – Hidden clone	-0.83	[-4.17, 2.50]	1.000	1.000
Llama3.1:8B	Proxy – No cover	-0.83	[-4.17, 2.50]	1.000	1.000
Llama3.1:8B	Read – No cover	1.67	[-2.50, 6.67]	0.727	1.000
Llama3.1:8B	Hidden clone – No cover	0.83	[-1.67, 3.33]	1.000	1.000
Llama3.1:8B	Live write – No cover	-55.00	[-62.50, -47.50]	2.71×10^{-20}	2.98×10^{-19}
Llama3.1:8B	Visible clone – No cover	3.33	[-4.17, 10.83]	0.523	1.000
Llama3.1:8B	Visible clone – Hidden clone	2.50	[-5.00, 10.00]	0.664	1.000

S2. Live TLS-record timing and planner-generated cover

This study tests cover on the network records produced by actual dispatch. It compares immediate controller cover, sequential timing matching, concurrent dispatch and a second planner that generates a complete cover execution. The source literature, the earlier receipt-level counterfactuals and this collection retain separate denominators.

S2.1. Population, protocol and execution

The study reuses the 300 executed-cover fixtures per model, with 120 training, 60 validation and 120 test fixtures, balanced over six requested actions. Qwen3:8B and Llama3.1:8B each execute all five arms, giving 3,000 formal episodes and 120 paired held-out fixtures per cell. Phrasings recur across splits. This is a new timing-layer collection on previously used workloads; earlier cover results were known when the protocol was designed.

LangGraph runs the same business planner against seven local MCP Streamable HTTP services over TLS 1.3/HTTP/2. Model weights and templates are fixed by layer hashes before and after collection. Model options are temperature zero, a 4,096-token context and at most 96 generated tokens per model step. One episode is active at a time; the local inference server has two execution slots so that the shadow chain can run alongside the real chain. Connections are shared by the two chains within an episode. A lock serializes calls to the same service, while calls to different services and planner inference may overlap. The experiment consequently exercises real concurrent chains without claiming unrestricted same-service multiplexing.

Every arm uses a common `/mcp` request path and 4,096/16,384-byte request/response body-padding quanta. The real planner receives only its own tool responses. Cover actions execute on isolated state and cannot directly write the real fixture. All requests and terminal responses are journaled; state outcomes are recomputed by replaying the synthetic operations. No episode is replaced after a missing response.

Five arms. *No cover* supplies the baseline. *Immediate* dispatches the original one or two cover calls directly after their triggering honest response. *Matched, sequential* waits for a sampled gap and holds the server response for a sampled minimum duration, blocking the real chain as required by that schedule. *Matched, concurrent* uses the same draws in a persistent episode-scoped dispatcher while the real planner continues inference. *Planner shadow* runs a complete second episode of the same model on an independently generated ticket, including its own checks, actions and feedback. It shares the service connections but not business state or planner context.

The shadow action is sampled uniformly from all six classes, including the real class. An independent fair draw selects which chain starts first; the later chain begins 0.2–1.5 s after the first ticket response. The attack is not told which first call is real. Random order removes that particular role cue, while the chain-separation attack tests whether other role information remains.

Historical timing pools. All no-cover episodes are collected first. Only their 120 training fixtures per model supply honest inter-call gaps and per-destination durations; the pools and collection amendments are frozen and pushed before any formal cover run. Timing-matched arms sample from those pools. A monotonic not-before wait enforces the selected gap, and the server holds a cover response until its selected minimum duration has elapsed. Transport, padding, scheduling and concurrent work still contribute to the achieved record times. The cover-arm order rotates across fixtures. Backend prompt caches persist between arms; a reused real ticket and a new shadow ticket can therefore have different cache histories.

S2.2. Observation and independent timing check

The primary observer receives destinations, bidirectional ciphertext counts, episode/call windows and relay timestamps at the completion of each encrypted TLS record. It receives no message body, operation, task identifier, ground-truth call role or model-link observation at test inference. Initialization precedes the call windows. TLS 1.3 outer type 23 identifies the encrypted records used to delimit a window; it is not interpreted as an application-plaintext type.

Npcap independently records both ingress directions. TCP sequence reassembly, separated by connection generation, must reproduce every relay record's outer type, length and count. The evaluator then measures relay-versus-packet completion-time error. Relay read completion is a software clock, not an exact NIC arrival timestamp. Supplied call boundaries and destination visibility remain observer privileges; the experiment does not evaluate discovery of those boundaries.

S2.3. Attacks, selection and prespecified decisions

Size-only candidates are standardized logistic regression, a 300-tree random forest, the frozen no-cover classifier, and last- and penultimate-destination rules. Timing candidates add direct timing-plus-size classification, supervised gap-detector removal, label-free two-component Gaussian-mixture gap removal, context-detector removal and joint decoding of zero, one or two removals. The context detector includes destinations, positions, sizes and within-episode timing relations. LR/RF choices and detector thresholds are selected on validation data; detector selection uses call-role F1 and then precision. Call-role annotations are permitted for training and evaluation, never as test-time inputs.

The shadow-specific attack partitions the observed calls into two causal chains with a beam of 64 candidates and a transition-gap model fitted on no-cover training traffic. A clean-chain classifier predicts each chain's action. A role selector, fitted on labelled shadow training episodes, chooses the real chain using observed chain summaries. It is not given the test roles. Exact partition recovery, real-chain

recovery and call-role accuracy are descriptive diagnostics. With perfect recovery of an unordered pair of independent uniform six-class labels, guessing the real label succeeds with probability $7/12$; this is an idealized pair-ambiguity reference, not a general privacy bound.

For each arm and model, validation accuracy selects a size-only attack and an adaptive attack from the complete suite; ties favor the declared size-first order. Test maxima are reported separately as descriptive results and do not replace the frozen choices. The test unit is the paired fixture, with 120 outcomes per cell.

Four testing families. Family P contains the six adaptive-minus-size comparisons for the sequential, concurrent and shadow arms. Each uses exact paired McNemar testing with Holm correction across the six comparisons. An arm meets the prespecified separability rule if either model gains at least 12/120 with adjusted $p < .05$. Family E tests the same six contrasts with paired t-TOST, margin $\pm 12/120$, 119 degrees of freedom and Holm correction. Both models must pass for an arm to meet the equivalence rule. At all ties, the implementation uses the conservative boundary probability $.9^{120}$ instead of zero. Equivalence concerns the validation-selected adaptive attack relative to the selected size attack; it does not establish cover-role indistinguishability or an all-attacker guarantee.

Family R has two immediate-arm replication comparisons: label-free removal against the selected size attack. Replication requires a gain of at least 24/120 and Holm-adjusted $p < .05$ in both models. Family C has eight size-only cover-versus-no-cover comparisons, also exact paired tests with Holm correction. C quantifies the size-view change; remaining adaptive inference is reported alongside it. Ten thousand action-stratified paired fixture bootstrap draws give intervals conditional on the fitted attacks and the reused workload. They omit refitting, attack-selection and workload-selection uncertainty. With 120 pairs, moderate increments or equivalence can remain unresolved.

S2.4. Costs and business outcomes

Real-chain latency begins at admission before either shadow chain starts and ends when the real planner finishes, including shadow-first waiting but excluding session initialization. Unique whole-connection bytes include setup and teardown without double-counting overlapping windows. Model calls and tokens include both planners in the shadow arm. Requested-goal success, whole-state correctness and honest-path agreement with no cover are reported on the same fixtures. Changes in planner behavior, cache history and shared inference resources accompany these policy comparisons, so an endpoint difference alone cannot isolate one causal mechanism.

S2.5. Complete attack and decision results

Tables S6–S7 give every frozen candidate; Table S8 gives every test, discordance count and conditional interval. Immediate removal meets R in both models: 112 versus 75 and 108 versus 80, with adjusted $p = 1.9 \times 10^{-8}$ and 4.3×10^{-6}. For historical sequential dispatch, P rejects for Llama (104 versus 82; $p = .00078$) but not Qwen (93 versus 76; $p = .0559$). Concurrent dispatch rejects for both (99 versus 66 and 106 versus 68). Each history-based arm consequently meets its prespecified separability rule through at least one model. The context attack uses non-timing inputs as well as time; these comparisons do not isolate the causal contribution of timestamps.

Both shadow selectors choose the same size attack for the size and adaptive roles, so their predictions are identical. Family E therefore passes by construction ($p_{\mathrm{Holm}} = 1.94 \times 10^{-5}$ under the all-tie safeguard). We classify this cell as degenerate: it is reported for completeness and is not counted as evidence that trace distributions or real/cover roles are indistinguishable. Section S2.6 gives the comparison that replaces it. Qwen's last-destination rule wins validation at 35/60, ahead of RF at 33/60, but on test scores 54/120 against RF's 70/120. Llama's selected RF scores 64/120. These selection differences are kept visible instead of choosing a new primary attack after test evaluation. All eight size-view cover-versus-none comparisons in C reject; the adaptive results show why a size-view decrease alone is an incomplete protection assessment.

Shadow roles and start order. The chain attack exactly partitions both chains in 48/120 Qwen and 59/120 Llama fixtures, but exactly selects the real chain in only 29 and 32. Its role assignments are correct on 563/977 and 673/1,255 calls; these calls are dependent within fixtures. Test data contain 53 shadow-first and 67 real-first fixtures per model, with 21 matching real/shadow action labels. The RF scores 28/53 and 42/67 for Qwen, and 25/53 and 39/67 for Llama, respectively. These descriptive strata expose order sensitivity without adding an inferential family. Shadow tasks themselves reach 116/120 and 89/120 requested goals. The fixed beam, role learner, recurring phrasings and different cache histories limit the tested attack suite.

S2.6. Ideal candidate-label reference for planner shadows

Let the real chain and k shadow chains carry labels $L_0, \ldots, L_k$ from C classes, with L_0 the real label. An idealized attacker observes the unordered multiset $\{L_0, \ldots, L_k\}$, which corresponds to perfect chain separation and per-chain classification, but not the index of the real chain. Under the idealization that the real chain is exchangeable with the shadows given the multiset (start order is randomized and the same planner runs both; reused real fixtures and their cache histories are the known departures), $\Pr[L_0 = c \mid \text{multiset}] = n_c/(k+1)$, where n_c counts chains labeled c, and the Bayes-optimal guess is a modal label. Its success probability is

$$P^*(k) = \frac{\mathbb{E}[\max_c n_c]}{k+1}.$$

For iid uniform shadow labels and $C = 6$, exact enumeration gives 1, $7/12$, $53/108$, $197/432$ and $185/432$ for $k =$

TABLE S6. ALL FROZEN QWEN3:8B ATTACKS ON LIVE-RECORD TEST OBSERVATIONS: CORRECT OUT OF 120 PER CELL. NONE IS NO COVER; HISTORY-BASED POLICIES SAMPLE MINIMUM GAP/RESPONSE HOLDS. DASHES INDICATE AN ATTACK NOT APPLIED TO NO-COVER TRAFFIC. VALIDATION CHOICES ARE REPORTED SEPARATELY FROM THESE INDIVIDUAL TEST SCORES.

Attack	None	Immediate	History seq.	History concurrent	Shadow
Size/destination LR	117	64	61	52	60
Size/destination RF	116	59	61	52	70
Frozen no-cover classifier	117	40	41	40	31
Last destination	114	75	76	66	54
Penultimate destination	14	43	42	47	43
Direct timing + size	104	60	57	47	64
Supervised gap removal	–	112	67	57	20
Label-free mixture removal	–	112	53	38	34
Context removal	–	111	93	87	20
Joint removal/task decoding	–	112	99	99	32
Two-chain separation	–	–	–	–	49

TABLE S7. ALL FROZEN LLAMA3.1:8B ATTACKS ON LIVE-RECORD TEST OBSERVATIONS: CORRECT OUT OF 120 PER CELL. NONE IS NO COVER; HISTORY-BASED POLICIES SAMPLE MINIMUM GAP/RESPONSE HOLDS. DASHES INDICATE AN ATTACK NOT APPLIED TO NO-COVER TRAFFIC. VALIDATION CHOICES ARE REPORTED SEPARATELY FROM THESE INDIVIDUAL TEST SCORES.

Attack	None	Immediate	History seq.	History concurrent	Shadow
Size/destination LR	110	82	79	68	59
Size/destination RF	112	80	82	68	64
Frozen no-cover classifier	110	54	57	57	43
Last destination	80	65	65	48	43
Penultimate destination	66	65	64	64	46
Direct timing + size	110	77	81	75	59
Supervised gap removal	–	108	100	98	20
Label-free mixture removal	–	108	14	66	43
Context removal	–	108	101	103	20
Joint removal/task decoding	–	108	104	106	33
Two-chain separation	–	–	–	–	34

$0, \ldots, 4$. If shadow labels are instead drawn without replacement from the $C-1$ classes other than the real one, every multiset has distinct labels, the posterior is uniform over the $k+1$ chains, and $P^*(k) = 1/(k+1)$ for $k \leq C-1$. This variant keeps each shadow's marginal label uniform. The script `experiment/shadow_floor.py` computes these values and writes `analysis/shadow-floor.json`.

On a realized split with m matching pairs among n fixtures and $k = 1$, the ideal reference is $(m+(n-m)/2)/n$. The live test split has $m = 21$ and $n = 120$, giving 70.5/120. The strongest tested call-window attack scores 70/120 (Qwen) and 64/120 (Llama), and the validation-selected attacks score 54 and 64, so the counts are compatible with this ideal reference. This is optimal accuracy for an ideal unordered-label observation, not a general upper or lower bound on trace inference. Actual traces may obscure labels or disclose roles. The 99 different-label fixtures provide a conditional task-label diagnostic with a one-half candidate-label reference. Aggregate counts restrict the forest's correct predictions on those fixtures only to $[49, 70]$ and $[43, 64]$. The exact task-label diagnostic requires the saved individual predictions, which remain local; it is distinct from direct real-chain role prediction.

S2.7. Achieved timing and the independent clock

Table S9 distinguishes sampled holds from achieved record durations. Each row includes all 455 cover calls in that arm/model, including training and validation. Sequential Qwen cover averages a .5198 s sampled gap and .5660 s achieved anchor gap; a .0405 s hold becomes a .1004 s observed duration. Concurrent duration increases to .1712 s. Llama's .0768 s mean hold becomes .1097 s sequential and .1234 s concurrent. Thus the live policy does not instantiate the exact timing-distribution transform used by the receipt-level counterfactual. Its measured inference, overhead and altered chronological context must be evaluated directly.

All 143,920 no-cover and 629,530 cover records match independently reassembled TCP/TLS records, across 21,000 connections and 19,244 call windows. All-record median absolute completion-time errors are 1.07 and 1.33 ms, with 99th percentiles 16.38 and 21.68 ms; the maximum is 1.675 s, including setup and teardown. Table S10 restricts the comparison to call endpoints: the maximum absolute error is 148.39 ms, and no episode's call order changes under the packet clock.

The separately labelled packet-clock diagnostic refits only the label-free mixture/removal calculation. Its task scores exactly reproduce the relay-clock scores in all eight

TABLE S8. ALL PRESPECIFIED LIVE-RECORD TEST FAMILIES. P/E COMPARE THE VALIDATION-SELECTED ADAPTIVE AND SIZE ATTACKS; R COMPARES LABEL-FREE IMMEDIATE REMOVAL WITH SIZE; C COMPARES EACH COVER SIZE ATTACK WITH NO COVER. +/− COUNT FIXTURE DISCORDANCES. INTERVALS ARE 95% PAIRED ACTION-STRATIFIED BOOTSTRAP INTERVALS CONDITIONAL ON THE FITTED ATTACKS. E USES PAIRED T-TOST WITH A ±10-POINT MARGIN; THE SHADOW SELECTORS CHOOSE THE IDENTICAL SIZE ATTACK, SO ALL PAIRS TIE. HOLM CORRECTION IS SEPARATE WITHIN EACH DECLARED FAMILY.

Family	Model	Arm	+/−	Δ (pp)	Conditional interval	Raw p	Holm p
P	Qwen	History seq.	32/15	14.2	[3.3, 25.0]	0.0186	0.0559
P	Qwen	History concurrent	42/9	27.5	[17.5, 38.3]	3.4×10^{-6}	1.7×10^{-5}
P	Qwen	Shadow	0/0	0.0	[0.0, 0.0]	1	1
P	Llama	History seq.	28/6	18.3	[10.0, 26.7]	2×10^{-4}	7.8×10^{-4}
P	Llama	History concurrent	41/3	31.7	[22.5, 40.8]	1.6×10^{-9}	9.7×10^{-9}
P	Llama	Shadow	0/0	0.0	[0.0, 0.0]	1	1
E	Qwen	History seq.	32/15	14.2	[3.3, 25.0]	0.7713	1
E	Qwen	History concurrent	42/9	27.5	[17.5, 38.3]	0.9992	1
E	Qwen	Shadow	0/0	0.0	[0.0, 0.0]	3.2×10^{-6}	1.9×10^{-5}
E	Llama	History seq.	28/6	18.3	[10.0, 26.7]	0.9643	1
E	Llama	History concurrent	41/3	31.7	[22.5, 40.8]	1	1
E	Llama	Shadow	0/0	0.0	[0.0, 0.0]	3.2×10^{-6}	1.9×10^{-5}
R	Qwen	Immediate	41/4	30.8	[21.7, 40.0]	9.3×10^{-9}	1.9×10^{-8}
R	Llama	Immediate	33/5	23.3	[15.0, 32.5]	4.3×10^{-6}	4.3×10^{-6}
C	Qwen	Immediate	0/42	-35.0	[-43.3, -26.7]	4.5×10^{-13}	2.7×10^{-12}
C	Qwen	History seq.	0/41	-34.2	[-42.5, -25.8]	9.1×10^{-13}	4.5×10^{-12}
C	Qwen	History concurrent	0/51	-42.5	[-50.8, -34.2]	8.9×10^{-16}	6.2×10^{-15}
C	Qwen	Shadow	0/63	-52.5	[-60.8, -43.3]	2.2×10^{-19}	1.7×10^{-18}
C	Llama	Immediate	3/33	-25.0	[-33.3, -16.7]	2.3×10^{-7}	4.5×10^{-7}
C	Llama	History seq.	2/30	-23.3	[-31.7, -15.0]	2.5×10^{-7}	4.5×10^{-7}
C	Llama	History concurrent	2/44	-35.0	[-44.2, -25.8]	3.1×10^{-11}	1.2×10^{-10}
C	Llama	Shadow	4/50	-38.3	[-47.5, -28.3]	3.8×10^{-11}	1.2×10^{-10}

cover cells, including 112 and 108 for immediate cover. On those held-out immediate calls it flags 181/182 and 182/182 decoys, with zero false flags among 490 and 634 honest calls. This is a sensitivity check for removal on supplied call windows; P/E remain relay-clock primary analyses, and the study makes no physical-network confidentiality claim.

S2.8. Resource costs, paths and business outcomes

Table S11 reports both the primary real-chain latency and a descriptive all-work duration through all planner chains and tool-response receipts. This prevents a concurrent dispatcher from hiding queued work behind real-planner completion. Mean all-work duration is 3.369 and 2.835 s for concurrent history dispatch, and 5.543 and 4.643 s for shadow execution. The corresponding no-cover means are 2.912 and 3.332 s. These are warm-session costs from this machine and execution order, not deployment throughput or a causal speedup estimate.

Whole-connection byte factors are 1.13–1.15 for controller policies and 1.31–1.35 for shadows; restricting accounting to supplied call windows gives 1.26–1.33 and 1.68–1.71. Shadow model-token factors are 2.01 and 1.99. Goal/state outcomes for shadow are 112/113 and 82/67 of 120, against no-cover 113/114 and 82/70. Honest paths exactly match no cover on 115/120 and 103/120 shadow fixtures, so even hidden isolated cover does not guarantee identical planning. No business-equivalence hypothesis is substituted for the prespecified accuracy equivalence test.

The formal dataset contains 3,000 primary episodes plus 600 shadow chains; every issued request has a terminal record and no unknown outcome was replaced. Active collector wall time is 38.8 min for no cover and 193.1 min for the four cover arms, including their service initialization and model loads but excluding the intervening pool/audit gate and later analyses. The artifact supplies implementations, frozen plans, timing pools and aggregate results. It supports arithmetic inspection but excludes individual record metadata and predictions, message bodies, captures, keys and weights. Feature reconstruction, refitting and independent per-fixture state replay require local inputs; their checks have separate receipts.

TABLE S9. ACHIEVED HISTORY-BASED COVER TIMING, ALL SPLITS (455 COVER CALLS PER ROW). MEANS IN SECONDS. AN ANCHOR GAP IS MEASURED FROM THE ACTUAL PRIOR COMPLETED ANCHOR CALL TO COVER RECORD START. SAMPLED DURATIONS ARE MINIMUM SERVER HOLDS, WHILE ACHIEVED DURATIONS INCLUDE RECORD-LEVEL TRANSPORT AND SCHEDULING.

Model	Arm	Drawn gap	Achieved anchor gap	Drawn hold	Achieved duration
Qwen	History seq.	0.5198	0.5660	0.0405	0.1004
Qwen	History concurrent	0.5198	0.5772	0.0405	0.1712
Llama	History seq.	0.5865	0.6203	0.0768	0.1097
Llama	History concurrent	0.5865	0.6204	0.0768	0.1234

TABLE S10. CALL-WINDOW CLOCK DIAGNOSTICS ACROSS ALL 300 FIXTURES PER CELL. RELAY VERSUS INDEPENDENT PACKET COMPLETION ERRORS, IN MILLISECONDS. VALUES ARE ABSOLUTE-ERROR MEDIAN / 99TH PERCENTILE / MAXIMUM. EVERY CALL RETAINS ITS CHRONOLOGICAL ORDER UNDER THE PACKET CLOCK. SETUP/TEARDOWN RECORDS ARE EXCLUDED FROM THIS TABLE.

Model	Arm	Call windows	Start: median / p99 / max	End: median / p99 / max
Qwen	None	1218	0.58 / 7.67 / 18.01	0.42 / 5.89 / 9.21
Qwen	Immediate	1673	1.70 / 8.57 / 32.76	1.23 / 4.20 / 7.98
Qwen	History seq.	1671	1.46 / 8.03 / 10.62	1.07 / 4.36 / 7.02
Qwen	History concurrent	1672	1.74 / 26.06 / 42.48	1.36 / 11.96 / 89.31
Qwen	Shadow	2411	2.17 / 57.62 / 148.39	1.72 / 25.46 / 116.16
Llama	None	1536	0.84 / 7.55 / 68.03	0.62 / 5.27 / 8.95
Llama	Immediate	1998	0.58 / 4.49 / 17.04	0.41 / 2.31 / 5.63
Llama	History seq.	1996	0.50 / 4.19 / 9.63	0.39 / 2.40 / 4.50
Llama	History concurrent	2002	0.58 / 5.54 / 71.35	0.43 / 6.65 / 29.04
Llama	Shadow	3067	0.72 / 12.10 / 99.22	0.64 / 9.71 / 89.08

TABLE S11. LIVE TEST-SET COSTS. REAL AND ALL ARE MEAN SECONDS FROM ADMISSION TO REAL-PLANNER FINISH AND TO ALL PLANNER/TOOL-RESPONSE COMPLETION, RESPECTIVELY; BOTH EXCLUDE SESSION SETUP/TEARDOWN. RECORD SPAN IS THE MEAN FIRST-TO-LAST TOOL-RECORD OBSERVATION. UNIQUE/CALL BYTES ARE SUMMED-BYTE FACTORS VERSUS NO COVER FOR COMPLETE CONNECTIONS AND SUPPLIED CALL WINDOWS. CALLS COUNT ALL TOOL/MODEL CALLS PER FIXTURE; TOKEN FACTORS INCLUDE THE SHADOW PLANNER. THESE ARE OBSERVED COSTS UNDER PERSISTENT CACHES AND PHASE ORDER, NOT ISOLATED CAUSAL SPEED MEASUREMENTS.

Model	Arm	Real s	All s	Span s	Unique×	Call×	Tool calls	Model calls	Tokens×
Qwen	None	2.912	2.912	2.272	1.00	1.00	4.08	4.08	1.00
Qwen	Immediate	3.575	3.575	2.899	1.15	1.33	5.60	4.08	1.00
Qwen	History seq.	4.280	4.280	3.611	1.15	1.32	5.59	4.08	1.00
Qwen	History concurrent	3.328	3.369	2.896	1.15	1.33	5.60	4.08	1.00
Qwen	Shadow	4.804	5.543	4.681	1.31	1.68	8.14	8.14	2.01
Llama	None	3.332	3.332	2.759	1.00	1.00	5.24	5.24	1.00
Llama	Immediate	2.916	2.916	2.420	1.13	1.27	6.80	5.28	1.01
Llama	History seq.	3.822	3.822	3.339	1.13	1.26	6.80	5.28	1.01
Llama	History concurrent	2.791	2.835	2.448	1.13	1.27	6.83	5.31	1.02
Llama	Shadow	3.939	4.643	4.087	1.35	1.71	10.46	10.45	1.99

S3. Record-only analysis without supplied call endpoints

This extension reuses all 3,000 main episodes. The analyst fixed its code and candidate grid before computing new test scores, with the older task outcomes already known. It is retrospective reused-workload evidence, not a new cohort. The underlying live collection, primary choices and tests remain unchanged.

S3.1. Inputs and observation scope

Each episode contributes seven final, closed connection logs, linked by the episode's connection bundle. All outer-type23 records are included; record time is made relative to the earliest such record. Initialization and closing records are retained. The feature whitelist is relative time, destination, direction and ciphertext record length. Labels and split identifiers join evaluation rows but never enter features. No per-call terminal, operation, private role or HTTP stream mapping is read by the attacker path. A source-hash manifest binds every input. Whole-task association is retained, so this is removal of call endpoints, not task localization.

S3.2. Candidate representations and fitting

Within each destination connection, records separated by more than 5, 20, 50, 100 or 250 ms form separate proposed events. Events retain the first and last timestamps and directional ciphertext sums. Destination summaries include event count, directional bytes, record count and maximum event volume; timing variants also include first/last times and total event duration. Ordered features use the first four and last sixteen events, with destination one-hot and log byte counts, adding duration and preceding gap in timing variants. Raw-record candidates treat each record as an event and use the same summaries. No plaintext or true call interval is required.

LR uses standardization, $C = 1$ and at most 3,000 iterations. RF has 300 trees. The fixed seed is 111121 and numerical libraries use two threads. Per-condition training fits each representation. A two-component mixture fits log gaps from that condition's training events without role labels. The crossing between its short/long components defines removal; the first event is retained as a heuristic. The remaining events go to a no-cover-trained task LR/RF at the same segmentation/view. Validation chooses baseline, removal and overall attacks by task accuracy with fixed-order ties. The selection file is written before that cell's test predictions.

There are forty-four fixed-grid candidates per model/condition: five gaps, two views, two learners for refitted baseline and clean-model removal, plus four raw-record candidates. Each cell retains 120 training, 60 validation and 120 test fixtures. The primary comparisons are selected removal minus selected baseline on immediate cover, with exact paired McNemar and Holm across the two models. Ten thousand paired test-fixture bootstrap draws give conditional intervals; they omit model selection, refitting and workload-selection uncertainty. Other arms and test maxima are descriptive. A test maximum is neither an independently selected confirmatory attack nor an upper bound on all possible inference.

S3.3. Exploratory protocol-aware eventization

After the Qwen idle-grid test results were known, a separate plan fixed one public-protocol alternative. An upward encrypted record of at least 4,096 bytes starts a proposed transaction. The first two such requests per connection correspond to the public MCP initialize/initialized setup sequence and are removed from this representation. A transaction ends at the last downward record of at least 4,096 bytes

before the next trigger, or at its start if no such response exists. This rule exploits the configured padding quantum and same-service serialization; it reads no true operation, call endpoint or private role. It is not an HTTP/2 plaintext parser or a general multiplexing solution.

The same size/timing views, LR/RF and clean-model GMM removal yield eight further candidates per cell. TRAIN/validation/test remain fixed, and validation selections are saved before their test predictions. These are exploratory results with no new superiority family. The first fixed-grid results and tests are preserved in full. The displayed Max spans all fifty-two candidates, while B/R still refer to the original fixed-grid validation choices. Complete candidate counts and predictions are in the local analysis package.

S3.4. Complete results

The independent boundary diagnostic is computed only after attack predictions, using saved private call truth. It is not an attacker feature or a segmentation selection criterion. The local package retains every candidate validation score, held-out prediction, input hash and reconstruction script. Individual record metadata and predictions remain local.

TABLE S12. SELECTED RECORD-ONLY ATTACKS. B/R/A REFER TO VALIDATION-SELECTED FIXED-GRID BASELINE, REMOVAL AND OVERALL ATTACKERS. PROTO B/R GIVES THE SEPARATE EXPLORATORY PROTOCOL-AWARE SELECTION; MAX SPANS BOTH SUITES. COUNTS ARE OUT OF 120. COMPLETE CANDIDATE OUTCOMES REMAIN IN THE LOCAL ANALYSIS PACKAGE.

Planner	Policy	B	R	Proto B/R	Max	Selected fixed-grid attacks (B; R; A)
Qwen	No cover	115	109	117/93	117	events 5ms LR; remove 20ms timing RF; events 5ms LR
Qwen	Immediate	80	69	68/91	91	records timing RF; remove 5ms RF; records timing RF
Qwen	History, sequential	73	74	66/45	75	records timing RF; remove 5ms RF; remove 5ms RF
Qwen	History, concurrent	56	61	51/47	62	events 5ms LR; remove 5ms RF; remove 5ms RF
Qwen	Planner shadow	69	65	66/52	73	events 20ms RF; remove 100ms timing RF; events 20ms RF
Llama	No cover	108	101	109/88	112	events 5ms timing LR; remove 20ms RF; events 5ms timing LR
Llama	Immediate	90	86	90/87	97	records RF; remove 5ms timing RF; records RF
Llama	History, sequential	88	56	90/71	93	events 5ms timing RF; remove 20ms LR; events 5ms timing RF
Llama	History, concurrent	78	62	83/68	88	events 50ms RF; remove 5ms LR; events 50ms RF
Llama	Planner shadow	69	56	68/52	69	events 5ms RF; remove 5ms RF; events 5ms RF

TABLE S13. PRESPECIFIED RECORD-ONLY IMMEDIATE-COVER COMPARISONS. DISCORDANCE COUNTS ARE REMOVAL-ONLY/BASELINE-ONLY CORRECT. INTERVALS CONDITION ON FITTED PREDICTORS AND FIXTURES.

Planner	Removal	Baseline	$+/-$	Difference pp	Conditional 95% interval	Raw p	Holm p
Qwen	69	80	3/14	-9.2	[-15.8, -2.5]	0.01273	0.02545
Llama	86	90	10/14	-3.3	[-11.7, 4.2]	0.5413	0.5413

TABLE S14. INDEPENDENT CALL-BOUNDARY AUDIT, IOU .5, AFTER PREDICTION. EACH ROW USES120 ASSOCIATED TEST EPISODES. IDLE SELECTS ITS BURST GAP BY VALIDATION TASK ACCURACY; PROTOCOL USES PUBLIC PADDING/SETUP RULES. VALUES ARE PRECISION / RECALL / F1, INCLUDING SPURIOUS PROPOSED EVENTS. THESE TRUTH COMPARISONS ARE NOT ATTACKER INPUTS.

Planner	Policy	Idle-gap eventization	Protocol-aware eventization
Qwen	No cover	0.000/0.000/0.000	0.233/0.434/0.303
Qwen	Immediate	0.000/0.000/0.000	0.229/0.412/0.295
Qwen	History sequential	0.000/0.000/0.000	0.272/0.489/0.350
Qwen	History concurrent	0.000/0.001/0.000	0.266/0.478/0.342
Qwen	Shadow	0.012/0.078/0.021	0.390/0.612/0.476
Llama	No cover	0.000/0.000/0.000	0.294/0.510/0.373
Llama	Immediate	0.000/0.000/0.000	0.302/0.512/0.380
Llama	History sequential	0.000/0.000/0.000	0.378/0.641/0.476
Llama	History concurrent	0.357/0.802/0.494	0.358/0.606/0.450
Llama	Shadow	0.000/0.000/0.000	0.428/0.637/0.512

S4. Reproduction scope and evidence archive

The accompanying artifact separates executable implementations, aggregate arithmetic and retained individual observations. The live-study implementation and frozen training pools identify the five policies. Aggregate attack tables give paired discordance counts and family membership for inspecting McNemar and Holm arithmetic. The record-only scripts retain their original candidate grid and explicitly separate the later exploratory eventizer.

Individual record metadata, predictions, message bodies, captures and TLS keys remain local. Full attack refitting, event reconstruction, per-fixture state replay and exact role diagnostics require those inputs. The artifact therefore supports code inspection and aggregate arithmetic, not independent full-data replication.

The evidence archive preserves the previous full supplement and its sources. It contains the background literature screening and source locators, coding and adjudication description, reporting examples, deployment maps, and the historical forwarding/concurrency pilot. Those measurements keep their original populations and uncertainty; the present main findings use the state control and live timing studies reported here. Artifact contents and limits are indexed in `ARTIFACT-INDEX.md`.

References

[1] Y. Zhang, X. Deng, Z. Gu, Y. Chen, K. Xu, Q. Li, and J. Wu, "Exposing LLM user privacy via traffic fingerprint analysis: A study of privacy risks in LLM agent interactions," *arXiv preprint arXiv:2510.07176*, 2025. [Online]. Available: https://arxiv.org/abs/2510.07176v1

[2] H. Jeong, M. Teymoorianfard, A. Kumar, A. Houmansadr, and E. Bagdasarian, "Network-level prompt and trait leakage in local research agents," in *35th USENIX Security Symposium*. USENIX Association, 2026, pp. 6485–6504. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/jeong

[3] B. Dangol, "From privacy to workflow integrity: Communication-graph metadata in autonomous agent interoperability," *arXiv preprint arXiv:2606.07150v3*, 2026. [Online]. Available: https://arxiv.org/abs/2606.07150v3

[4] S. Pouryousef, J. Lopez, S. R. Nowmi, M. M. Kamol, M. Hossain, M. Tran, and M. S. Rahman, "A large-scale benchmark and risk assessment of traffic analysis attacks on cloud LLM services," *arXiv preprint arXiv:2609.31877v2*, 2026. [Online]. Available: https://arxiv.org/abs/2609.31877v2

[5] C. Song, H. Wang, and R. Martin, "PersonaFingerprint: Measuring persona inference on modern websites with LLM-driven browsing," *arXiv preprint arXiv:2605.15962v1*, 2026. [Online]. Available: https://arxiv.org/abs/2605.15962v1

[6] D. Kang, H. Jeong, J. Sheffey, P. Datta, and A. Houmansadr, "Whose agent are you? multi-layer fingerprinting and attribution of autonomous web agents," *arXiv preprint arXiv:2606.20910v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.20910v1

[7] I. Fayolle, S. Bouhenniche, S. Pélissier, P. Laperdrix, C. Maurice, and W. Rudametkin, "On the internet, nobody knows you're an LLM bot: Unmasking web agents with multi-layer fingerprinting," *arXiv preprint arXiv:2606.30119v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.30119v1

[8] A. Shahriar, M. N. Rahman, S. Ahmed, F. Sadeque, and M. R. Parvez, "AgentTell: Behavioural side-channel leakage in browser-use agents," *arXiv preprint arXiv:2609.32915v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.32915v1

[9] F. Li, X. Gao, Z. Liu, T. Li, C. Fu, Z. Wang, and K. Xu, "ANT: A multi-granularity network traffic dataset and benchmark for agents behavior auditing," *arXiv preprint arXiv:2610.06514v1*, 2026. [Online]. Available: https://arxiv.org/abs/2610.06514v1

[10] D. Ayzenshteyn, R. Weiss, and Y. Mirsky, "Cloak, honey, trap: Proactive defenses against LLM agents," in *34th USENIX Security Symposium*. USENIX Association, 2025, pp. 8095–8114. [Online]. Available: https://www.usenix.org/conference/usenixsecurity25/presentation/ayzenshteyn

[11] L. Song, Z. Pang, W. Wang, Z. Wang, X. Wang, H. Chen, W. Song, Y. Jin, D. Meng, and R. Hou, "The early bird catches the leak: Unveiling timing side channels in LLM serving systems," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 11 431–11 446, 2025, cited version: arXiv:2409.20002v5. [Online]. Available: https://ieeexplore.ieee.org/document/11206380/

[12] G. McDonald and J. B. Or, "Whisper leak: a side-channel attack on large language models," *arXiv preprint arXiv:2511.03675*, 2025, cited version: arXiv:2511.03675v1. [Online]. Available: https://arxiv.org/abs/2511.03675v1

[13] T. Zhang, G. Saileshwar, and D. Lie, "Time will tell: Timing side channels via output token count in large language models," *arXiv preprint arXiv:2412.15431*, 2024, cited version: arXiv:2412.15431v1. [Online]. Available: https://arxiv.org/abs/2412.15431v1

[14] G. Wu, Z. Li, Y. Zhang, Z. Zhang, J. Niu, Y. Wu, and Y. Zhang, "Cacheprune: Privacy-aware and fine-grained KV cache sharing for efficient LLM inference," *arXiv preprint arXiv:2605.23640v1*, 2026. [Online]. Available: https://arxiv.org/html/2605.23640v1

[15] M. Juarez, S. Afroz, G. Acar, C. Diaz, and R. Greenstadt, "A critical evaluation of website fingerprinting attacks," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2014, pp. 263–274.

[16] N. Mathews, J. K. Holland, S. E. Oh, M. S. Rahman, N. Hopper, and M. Wright, "SoK: A critical evaluation of efficient website fingerprinting defenses," in *IEEE Symposium on Security and Privacy*, 2023.

[17] T. Wang and I. Goldberg, "Walkie-talkie: An efficient defense against passive website fingerprinting attacks," in *26th USENIX Security Symposium*, 2017. [Online]. Available: https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/wang-tao

[18] J. Gong and T. Wang, "Zero-delay lightweight defenses against website fingerprinting," in *29th USENIX Security Symposium*. USENIX Association, 2020, pp. 717–734. [Online]. Available: https://www.usenix.org/conference/usenixsecurity20/presentation/gong

[19] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Advances in Neural Information Processing Systems*, vol. 33, 2020. [Online]. Available: https://arxiv.org/abs/2002.08347v2

[20] R. Weiss, D. Ayzenshteyn, G. Amit, and Y. Mirsky, "What was your prompt? a remote keylogging attack on ai assistants," in *33rd USENIX Security Symposium*, 2024. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/weiss

[21] N. Carlini and M. Nasr, "Remote timing attacks on efficient language model inference," *arXiv preprint arXiv:2410.17175v1*, 2024. [Online]. Available: https://arxiv.org/html/2410.17175v1

[22] D. Ahmed, A. Sabir, and A. Das, "Spying through your voice assistants: Realistic voice command fingerprinting," in *32nd USENIX Security Symposium*, 2023, pp. 2419–2436. [Online]. Available: https://www.usenix.org/conference/usenixsecurity23/presentation/ahmed-dilawer

[23] Model Context Protocol, "Transports," 2025, protocol version 2025-06-18. [Online]. Available: https://modelcontextprotocol.io/specification/2025-06-18/basic/transports

[24] M. Thomson and C. Benfield, "HTTP/2," RFC 9113, 2022. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9113.html

[25] E. Rescorla, "The transport layer security (tls) protocol version 1.3," RFC 8446, 2018. [Online]. Available: https://www.rfc-editor.org/rfc/rfc8446.html

[26] W. Lugoloobi, S. Marro, J. Magomere, J. Wright, and C. Russell, "Known by their actions: Fingerprinting LLM browser agents via UI traces," *arXiv preprint arXiv:2605.14786v1*, 2026. [Online]. Available: https://arxiv.org/abs/2605.14786v1

[27] J. Roh, E. Bagdasarian, H. Haddadi, and A. S. Shamsabadi, "SPILLage: Agentic oversharing on the web," *arXiv preprint arXiv:2602.13516v1*, 2026. [Online]. Available: https://arxiv.org/abs/2602.13516v1

[28] S. Chen, R. Wang, X. Wang, and K. Zhang, "Side-channel leaks in web applications: A reality today, a challenge tomorrow," in *IEEE Symposium on Security and Privacy*, 2010. [Online]. Available: https://staff.ie.cuhk.edu.hk/~khzhang/my-papers/2010-oakland-side-channel.pdf

[29] P. Chapman and D. Evans, "Automated black-box detection of side-channel vulnerabilities in web applications," in *ACM Conference on Computer and Communications Security*, 2011. [Online]. Available: https://www.cs.virginia.edu/~evans/pubs/ccs2011/

[30] P. Reynolds, J. L. Wiener, J. C. Mogul, M. K. Aguilera, and A. Vahdat, "WAP5: Black-box performance debugging for wide-area systems," in *International World Wide Web Conference*, 2006. [Online]. Available: https://cseweb.ucsd.edu/~vahdat/papers/wap5-www06.pdf

[31] B. C. Tak, C. Tang, C. Zhang, S. Govindan, B. Urgaonkar, and R. N. Chang, "vPath: Precise discovery of request processing paths from black-box observations of thread and network activities," in *USENIX Annual Technical Conference*, 2009. [Online]. Available: https://www.usenix.org/conference/usenix-09/vpath-precise-discovery-request-processing-paths-black-box-observations-thread

[32] K. P. Dyer, S. E. Coull, T. Ristenpart, and T. Shrimpton, "Peek-a-boo, i still see you: Why efficient traffic analysis countermeasures fail," in *IEEE Symposium on Security and Privacy*, 2012. [Online]. Available: https://pages.cs.wisc.edu/~rist/papers/trafanal.pdf

[33] R. Trimananda, J. Varmarken, A. Markopoulou, and B. Demsky, "Packet-level signatures for smart home devices," in *Network and Distributed System Security Symposium*, 2020. [Online]. Available: https://www.ndss-symposium.org/ndss-paper/packet-level-signatures-for-smart-home-devices/

[34] N. Apthorpe, D. Y. Huang, D. Reisman, A. Narayanan, and N. Feamster, "Keeping the smart home private with smart(er) IoT traffic shaping," *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 3, pp. 128–148, 2019. [Online]. Available: https://petsymposium.org/popets/2019/popets-2019-0040.pdf

[35] D. Duan, Q. Liu, Y. Liu, N. Zou, X. Hu, and D. Liu, "How much can tool calls reveal about you? privacy leakage from tool-call traces," *arXiv preprint arXiv:2610.41885v1*, 2026. [Online]. Available: https://arxiv.org/abs/2610.41885v1

[36] Y. Yao, Z. Wang, H. Cheng, Y. Cheng, H. Du, and X.-Y. Li, "IntentMiner: Intent inversion attack via tool call analysis in the model context protocol," *arXiv preprint arXiv:2512.14166v2*, 2026. [Online]. Available: https://arxiv.org/abs/2512.14166v2

[37] P. Sirinam, M. Imani, M. Juarez, and M. Wright, "Deep fingerprinting: Undermining website fingerprinting defenses with deep learning," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2018. [Online]. Available: https://arxiv.org/abs/1801.02265v5

[38] S. Bhat, D. Lu, A. Kwon, and S. Devadas, "Var-CNN: A data-efficient website fingerprinting attack based on deep learning," *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 4, pp. 292–310, 2019. [Online]. Available: https://petsymposium.org/popets/2019/popets-2019-0070.php

[39] M. Bishop, "HTTP/3," RFC 9114, 2022. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9114.html

[40] J. Iyengar and M. Thomson, "QUIC: A UDP-based multiplexed and secure transport," RFC 9000, 2021. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9000.html

[41] C. Song, D. D. M. Mekala, H. Wang, and R. P. Martin, "Redefining website fingerprinting attacks with multi-agent LLMs," *Proceedings on Privacy Enhancing Technologies*, vol. 2026, no. 3, pp. 688–702, 2026. [Online]. Available: https://www.petsymposium.org/popets/2026/popets-2026-0101.php

[42] S. Zavrak, "Content-aware attack detection in LLM agent tool-call traffic: An empirical study of features, architectures, and evaluation protocols," *arXiv preprint arXiv:2605.11053v3*, 2026. [Online]. Available: https://arxiv.org/abs/2605.11053v3

[43] Z. Guan, G. Xiong, G. Gou, Z. Li, M. Cui, and C. Liu, "BAPM: Block attention profiling model for multi-tab website fingerprinting attacks on Tor," in *Annual Computer Security Applications Conference*, 2021.

[44] X. Deng, Q. Yin, Z. Liu, X. Zhao, Q. Li, M. Xu, K. Xu, and J. Wu, "Robust multi-tab website fingerprinting attacks in the wild," in *IEEE Symposium on Security and Privacy*, 2023, pp. 1005–1022.

[45] T. Wang and I. Goldberg, "On realistically attacking Tor with website fingerprinting," *Proceedings on Privacy Enhancing Technologies*, vol. 2016, no. 4, pp. 21–36, 2016.

[46] J. Hayes and G. Danezis, "k-fingerprinting: A robust scalable website fingerprinting technique," in *25th USENIX Security Symposium*, 2016, pp. 1187–1203. [Online]. Available: https://www.usenix.org/conference/usenixsecurity16/technical-sessions/presentation/hayes

[47] G. Zhao, H. Wu, and X. Zhang, "AgentTrace: Execution-level fingerprinting of black-box LLM agents," 2026, version 1. [Online]. Available: https://arxiv.org/abs/2610.32092v1

[48] M. Ediga and S. Chattopadhyay, "Trace: Unmasking AI attack agents through terminal behavior fingerprinting," 2026, version 1. [Online]. Available: https://arxiv.org/abs/2605.01186v1

[49] O. Borysenko, "Developer experience with AI coding agents: HTTP behavioral signatures in documentation portals," 2026, extended version 3. [Online]. Available: https://arxiv.org/abs/2604.02544v3

[50] ——, "Documentation retrieval by AI coding agents: An empirical study of HTTP behavioral signatures," in *Proceedings of the 1st International Workshop on Agentic AI for Next-Generation Software Development*. ACM, 2026, pp. 168–169.

[51] Reworr and D. Volkov, "LLM agent honeypot: Monitoring AI hacking agents in the wild," 2025, version 2; first submitted 2024. [Online]. Available: https://arxiv.org/abs/2410.13919v2

[52] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in *31st USENIX Security Symposium (USENIX Security 22)*. USENIX Association, 2022, pp. 3971–3988. [Online]. Available: https://www.usenix.org/conference/usenixsecurity22/presentation/arp

[53] Y. Yao, J. Duan, K. Xu, Y. Cai, Z. Sun, and Y. Zhang, "A survey on large language model (LLM) security and privacy: The good, the bad, and the ugly," *arXiv preprint arXiv:2312.02003v3*, 2024. [Online]. Available: https://arxiv.org/abs/2312.02003v3

[54] A. Shahriar, M. N. Rahman, S. Ahmed, F. Sadeque, and M. R. Parvez, "A survey on agentic security: Applications, threats and defenses," *arXiv preprint arXiv:2510.06445v1*, 2025. [Online]. Available: https://arxiv.org/abs/2510.06445v1

[55] M. A. Ferrag, N. Tihanyi, D. Hamouda, L. Maglaras, and M. Debbah, "From prompt injections to protocol exploits: Threats in LLM-powered AI agents workflows," *arXiv preprint arXiv:2506.23260v1*, 2025. [Online]. Available: https://arxiv.org/abs/2506.23260v1

[56] LangChain, "LangGraph, version 1.0.3," Software release, November 10, 2025. [Online]. Available: https://pypi.org/project/langgraph/1.0.3/

[57] E. Debenedetti, I. Shumailov, T. Fan, J. Hayes, N. Carlini, D. Fabian, C. Kern, C. Shi, A. Terzis, and F. Tramèr, "Defeating prompt injections by design," 2025, version 2. [Online]. Available: https://arxiv.org/abs/2503.18813v2

[58] M. Costa, B. Köpf, A. Kolluri, A. Paverd, M. Russinovich, A. Salem, S. Tople, L. Wutschitz, and S. Zanella-Béguelin, "Securing AI agents with information-flow control," 2025, version 2. [Online]. Available: https://arxiv.org/abs/2505.23643v2

[59] T. Shi, J. He, Z. Wang, H. Li, L. Wu, W. Guo, and D. Song, "Progent: Programmable privilege control for LLM agents," 2025, version 2. [Online]. Available: https://arxiv.org/abs/2504.11703v2

[60] S. Yao, N. Shinn, P. Razavi, and K. Narasimhan, "τ-bench: A benchmark for tool-agent-user interaction in real-world domains," 2024, version 1. [Online]. Available: https://arxiv.org/abs/2406.12045v1

[61] E. Debenedetti, J. Zhang, M. Balunović, L. Beurer-Kellner, M. Fischer, and F. Tramèr, "AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents," in *Advances in Neural Information Processing Systems*, vol. 37, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/97091a5177d8dc64b1da8bf3e1f6fb54-Abstract-Datasets_and_Benchmarks_Track.html

[62] X. Cai, R. Nithyanand, T. Wang, R. Johnson, and I. Goldberg, "A systematic approach to developing and evaluating website fingerprinting defenses," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2014, pp. 227–238.

[63] D. C. Howe and H. Nissenbaum, "TrackMeNot: Resisting surveillance in web search," in *Lessons from the Identity Trail: Anonymity, Privacy, and Identity in a Networked Society*. Oxford University Press, 2008, pp. 417–436.

[64] S. T. Peddinti and N. Saxena, "On the privacy of web search based on query obfuscation: A case study of TrackMeNot," in *Privacy Enhancing Technologies (PETS)*, ser. LNCS, vol. 6205. Springer, 2010, pp. 19–37.

[65] E. Balsa, C. Troncoso, and C. Diaz, "OB-PWS: Obfuscation-based private web search," in *IEEE Symposium on Security and Privacy*, 2012, pp. 491–505.

[66] M. Juarez, M. Imani, M. Perry, C. Diaz, and M. Wright, "Toward an efficient website fingerprinting defense," in *European Symposium on Research in Computer Security (ESORICS)*, ser. LNCS, vol. 9878. Springer, 2016, pp. 27–46.

[67] J. Domingo-Ferrer, A. Solanas, and J. Castellà-Roca, "h(k)-private information retrieval from privacy-uncooperative queryable databases," *Online Information Review*, vol. 33, no. 4, pp. 720–744, 2009.

[68] A. Gervais, R. Shokri, A. Singla, S. Capkun, and V. Lenders, "Quantifying web-search privacy," in *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2014, pp. 966–977.

[69] V. Shmatikov and M.-H. Wang, "Timing analysis in low-latency mix networks: Attacks and defenses," in *European Symposium on Research in Computer Security (ESORICS)*, ser. LNCS, vol. 4189. Springer, 2006, pp. 18–33.